Monte-Carlo Methods
Chapter 5

- Assume model information is not known, i.e., $p(s', r | s, a)$ are not available

- However, sample trajectories can be observed, i.e., starting from any initial state s_0, we can observe trajectory until termination.

- Data available on an episode:

$$\underbrace{S_0, A_0, R_1, S_1}_{1^{st}\text{ data tuple}}, \underbrace{A_1, R_2, S_2}_{2^{nd}\text{ data tuple}}, \ldots, \underbrace{S_{T-1}, A_{T-1}, R_T, S_T}_{\text{final data tuple.}}$$

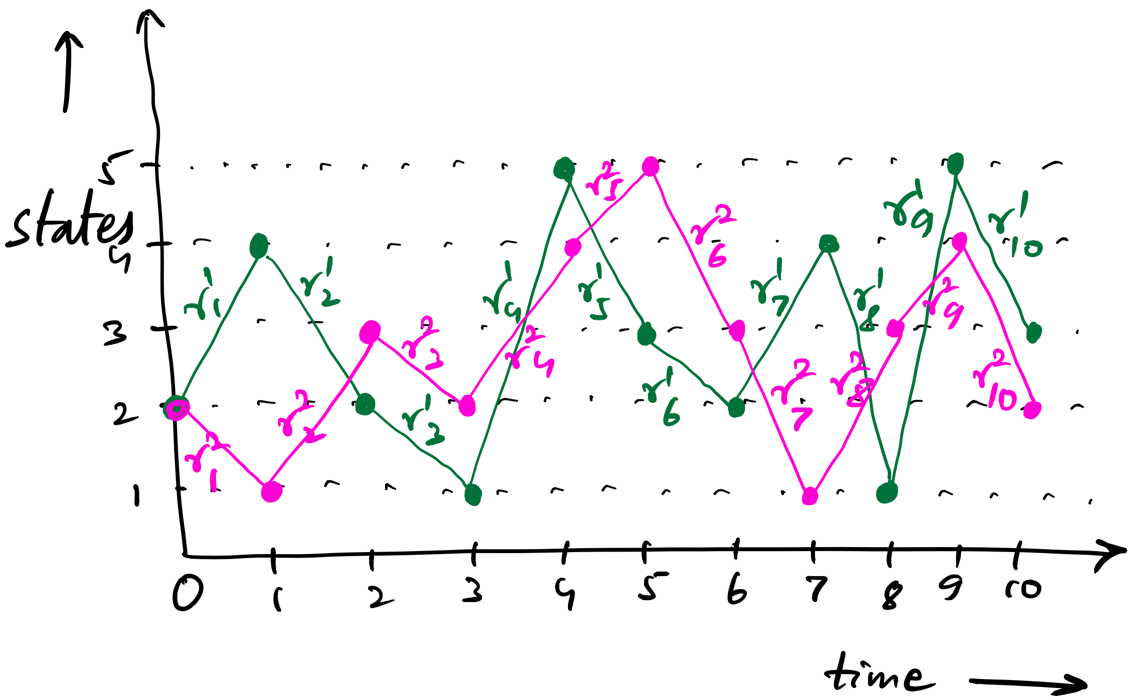
- If task is episodic
  - T is termination time of task
- If task is continuing
  - T is a fixed time set by decision maker for trajectory length

- Since $p(s', r \mid s, a)$ are not known, we cannot directly compute

$$v_\pi(s) = E_\pi\left[G_t \mid S_t = s\right]$$

$$= E_\pi\left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid S_t = s\right]$$

Monte - Carlo Prediction

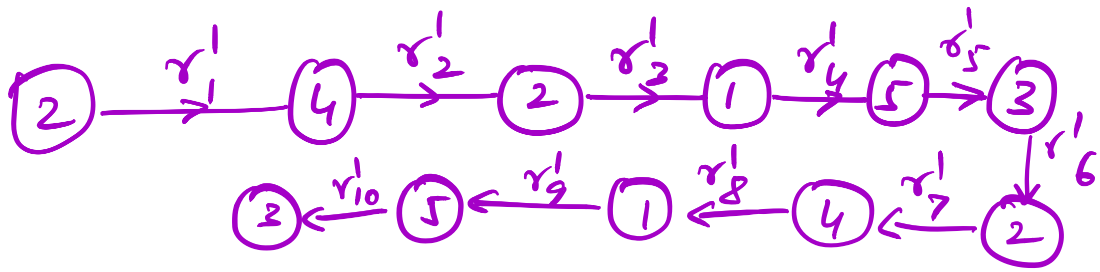


Two approaches for collecting sample estimates:

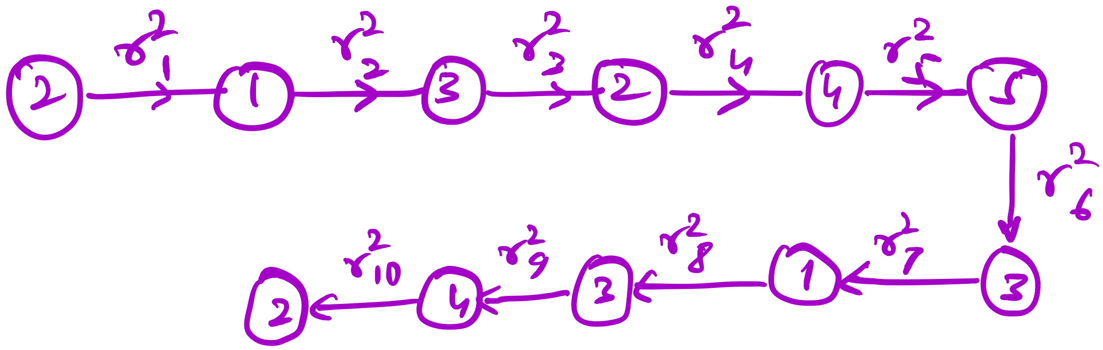
(i) **First visit Method:**

Trajectories for Example above:

a)



b)



Sample estimates of values:

$$\hat{v}(2) := r^1_1 + \gamma r^1_2 + \gamma^2 r^1_3 + \cdots + \gamma^9 r^1_{10}$$

$$\hat{v}(4) := r^1_2 + \gamma r^1_3 + \gamma^2 r^1_4 + \cdots + \gamma^8 r^1_{10}$$

$$\hat{v}(1) := r^1_4 + \gamma r^1_5 + \cdots + \gamma^6 r^1_{10}$$

$$\hat{v}(5) := r^1_5 + \gamma r^1_6 + \cdots + \gamma^5 r^1_{10}$$

$$\hat{v}(3) := r_6^1 + \gamma r_7^1 + \cdots + \gamma^4 r_{10}^1$$

$$\hat{v}(2) := r_1^2 + \gamma r_2^2 + \cdots + \gamma^9 r_{10}^2$$

$$\hat{v}(1) := r_2^2 + \gamma r_3^2 + \cdots + \gamma^8 r_{10}^2$$

$\vdots$

(ii) **Every-Visit Method:**

Additional samples available

$$\hat{v}(2) := r_3^1 + \gamma r_4^1 + \cdots + \gamma^7 r_{10}^1$$

$$\hat{v}(2) := r_7^1 + \gamma r_8^1 + \gamma^2 r_9^1 + \gamma^3 r_{10}^1$$

$\vdots$

Similarly for the 2nd trajectory.

$\vdots$

Important Remarks:

(i) First visit MC method estimates $V_\pi(s)$ as the average of returns following first visits to state s

(ii) Every visit MC method estimates $V_\pi(s)$ as the average of returns following every visit to state s

(iii) Both first and every visit M.C methods converge to $V_\pi(s)$ as the number of visits to s tend to infinity

<u>Algorithm</u>: First visit MC Prediction for Estimating $V \approx V_\pi$

Input: a policy π to be evaluated

Initialize: $V(s) \in \mathbb{R}$, arbitrarily, $\forall s \in S$

Returns$(s) \leftarrow$ empty list, $\forall s \in S$

Loop forever (for each episode):

- Generate an episode following π:

  $S_0, A_0, R_1, S_1, A_1, R_2, S_2, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$

- $G \leftarrow 0$

- Loop for each step of episode $t = T-1, T-2, \ldots, 0$:

  $G \leftarrow \gamma G + R_{t+1}$

  Unless S_t appears in $S_0, S_1, \ldots, S_{t-1}$

  Append G to Returns(S_t)

$$V(S_t) \leftarrow \text{average}(\text{Returns}(S_t))$$

* Every visit MC is same as First visit MC except for the condition

Unless S_t appears in $S_0, S_1, \dots, S_{t-1}$

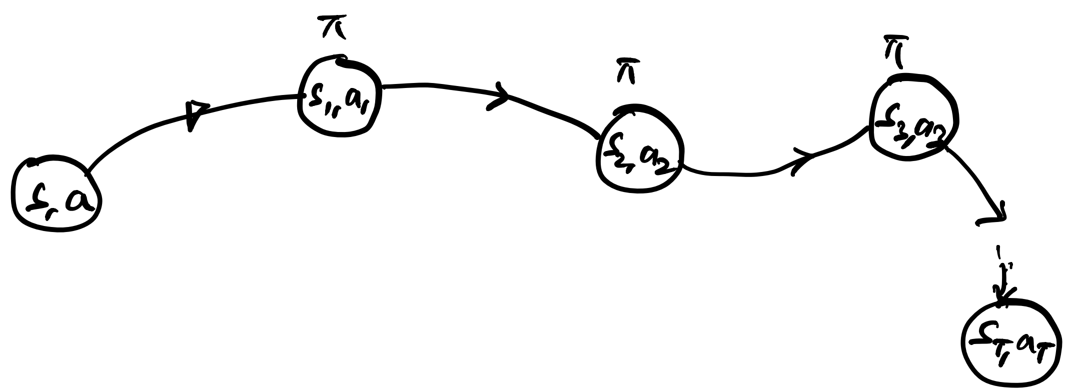
above

Monte-Carlo Estimation of $q_\pi(s,a)$

- Start from a given (s,a) tuple and pick subsequent actions according to policy π
- We say that a state-action pair $(\bar{s}, \bar{a})$ is visited if state $\bar{s}$ is

visited and action $\bar{a}$ is chosen in it

- First visit & every visit methods may now be defined as before



Problem: - When π is a deterministic policy, then in any state $\hat{s}$, a given action $\hat{a} = \pi(\hat{s})$ is always chosen.

- We won't get estimates of many (s, a) tuples.

Ways around:

(i) Choose starting (s, a) tuple randomly s.t. each (s, a) tuple has a positive probability of getting selected. This is called **Assumption of exploring starts.**

(ii) Use stochastic policies π to generate actions. Here $\pi(\bar{a}|\bar{s}) > 0 \;\forall\; \bar{a} \in A(\bar{s})$.

Monte-Carlo Control:

Goal: Find the optimal policy π_* using Monte-Carlo techniques

Algorithm: Monte-Carlo ES (Exploring starts) for Estimating $\pi \approx \pi_*$

Initialize: $\pi(s) \in A(s)$ (arbitrarily), $\forall s \in S$

$Q(s,a) \in \mathbb{R}$ (arbitrarily), $\forall s \in S,\ a \in A(s)$

$Returns(s,a) \leftarrow$ empty list, $\forall s \in S,\ a \in A(s)$

Loop forever (for each episode):

- Choose $S_0 \in S$, $A_0 \in A(S_0)$ randomly s.t. all pairs have probability > 0

- Generate an episode starting from (S_0, A_0) following π:

$S_0, A_0, R_1, S_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$

- $G \leftarrow 0$
- Loop for each step of episode $t = T-1, T-2, \ldots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair (S_t, A_t) appears in $(S_0, A_0), (S_1, A_1), \ldots, (S_{t-1}, A_{t-1})$:

- Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow average(Returns(S_t, A_t))$

$\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$

Alternatives to Exploring Starts

Approaches

- On-policy methods → Evaluate/Improve policy used to generate data
- Off-policy methods → Evaluate/Improve policy different from that used to generate data

• Example of on-policy control policies:

ϵ- greedy policies

$$- P(\text{greedy action}) = 1 - \epsilon + \frac{\epsilon}{|A(s)|}$$

- $P(\text{nongreedy action}) = \dfrac{\epsilon}{|A(s)|}$

Algorithm: On-policy first-visit MC Control (for ϵ-soft policies), estimating $\pi \approx \pi_*$

- Algorithm parameter: small $\epsilon > 0$
- Initialize:

  $\pi \leftarrow$ an arbitrary ϵ-soft policy

  $Q(s,a) \in \mathbb{R}$ (arbitrary values), $\forall s \in S,\ a \in A(s)$

  $Returns(s,a) \leftarrow$ empty list $\forall s \in S,\ a \in A(s)$

- Repeat forever (for each episode):
  - Generate an episode following π:

    $S_0, A_0, R_1, S_1, A_1, R_2, S_2, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$

- $G \leftarrow 0$
- Loop for each step of episode $t = T-1, T-2, \dots, 0$
  - $G \leftarrow \gamma G + R_{t+1}$
  - Unless the pair (S_t, A_t) appears in $(S_0, A_0), (S_1, A_1), \dots, (S_{t-1}, A_{t-1})$:

    Append G to $Returns(S_t, A_t)$

    $Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

    $A^* \leftarrow \underset{a}{\arg\max}\ Q(S_t, a)$

    For all $a \in A(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1-\epsilon+\frac{\epsilon}{|A(S_t)|} & \text{if } a = A^* \\ \frac{\epsilon}{|A(S_t)|} & \text{if } a \neq A^* \end{cases}$$

Off-Policy Methods for Prediction

Problem: Data is generated according to a policy b (the behavior policy) while the value v_π of another policy (the target policy) needs to be predicted

On-policy Methods	Off-policy Methods
1. Work with data generated from target policy itself	1. Work with data generated from behavior policy

2. Lower variance and faster convergence	2. Higher variance and slower convergence (since data is from a different policy)
3. Less general methods	3. More general and powerful approaches as they contain on-policy methods for the case where target policy = behavior policy

Assume –

- target policy is π
- behavior policy is b
- both π & b are fixed policies

- **Key Assumption**

If $\pi(a|s) > 0$ for some action a,

then $b(a|s) > 0$ as well

[Assumption of coverage : support of b must at least contain the support of π]

Problem: Off-policy prediction

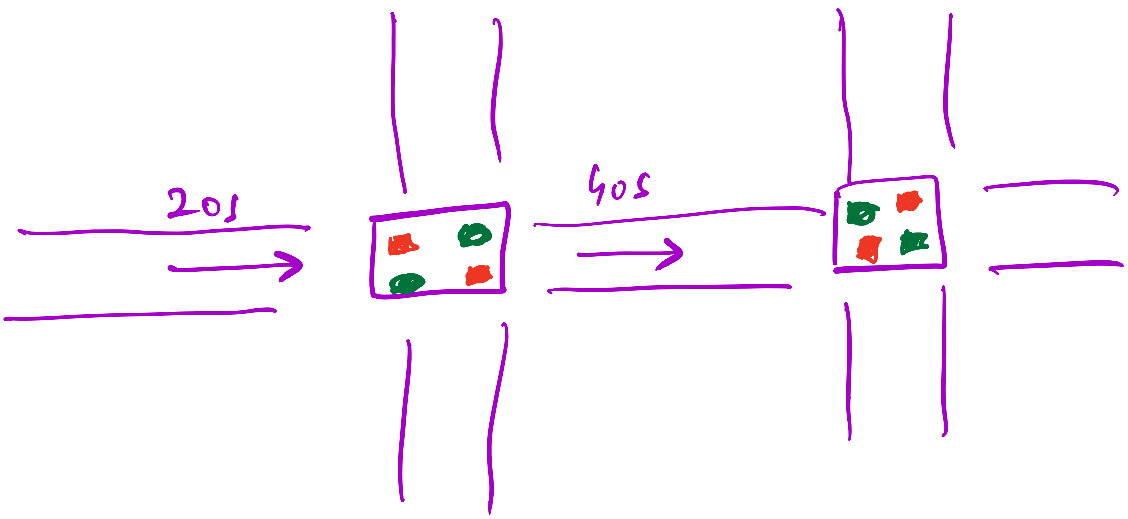
- Estimate v_π or q_π but with episodes obtained from policy b (behavior policy)

- **Solution methodology used :**

Importance sampling

↳ Estimates expected values under one distribution given samples from another.

Example – Vehicular traffic Control



Problem: Optimal signal Control

– how much green time to give

for a particular lane given a certain level of congestion.

Off-policy Methods

Recall π : target policy

b : behavior policy

$$P(A_t, S_{t+1}, A_{t+1}, \ldots, S_T \mid S_t, A_{t:T-1} \sim \pi)$$

$$= P(S_T \mid S_{T-1}, A_{T-1}, \ldots, S_{t+1}, A_t, S_t, A_{t:T-1} \sim \pi)$$

$$* P(S_{T-1}, A_{T-1}, \ldots, S_{t+1}, A_t \mid S_t, A_{t:T-1} \sim \pi)$$

$$\|$$

$$P(A_{T-1} \mid S_{T-1}, S_{T-2}, A_{T-2}, \dots, S_{t+1}, S_t, A_{t:T-1} \sim \pi)$$

$$* \; P(S_{T-1}, S_{T-2}, \dots, S_{t+1}, S_t, A_{t:T-1} \sim \pi)$$

$$\|$$

$$P(S_{T-1} \mid S_{T-2}, A_{T-2}, \dots, S_{t+1}, S_t, A_{t:T-1} \sim \pi)$$

$$* \; P(S_{T-2}, \dots, S_t, A_{t:T-1} \sim \pi)$$

$$= P(S_T \mid S_{T-1}, A_{T-1})\, \pi(A_{T-1} \mid S_{T-1})$$

$$* \; P(S_{T-1} \mid S_{T-2}, A_{T-2})\, \pi(A_{T-2} \mid S_{T-2})$$

$$\cdots \; P(S_{t+1} \mid S_t, A_t)\, \pi(A_t \mid S_t)$$

$$= \prod_{k=t}^{T-1} \pi(A_k \mid S_k)\, p(S_{k+1} \mid S_k, A_k)$$

transition probability of going to state S_{k+1} from state S_k when action chosen is A_k

Similarly,

$$P(A_t, S_{t+1}, A_{t+1}, \ldots, S_T \mid S_t, A_{t:T-1} \sim b)$$

$$= \prod_{k=t}^{T-1} b(A_k \mid S_k)\, p(S_{k+1} \mid S_k, A_k)$$

Define Importance Sampling (IS) ratio as:

$$\rho_{t:T-1} = \frac{P(A_t, S_{t+1}, \ldots, S_T \mid S_t, A_{t:T-1} \sim \pi)}{P(A_t, S_{t+1}, \ldots, S_T \mid S_t, A_{t:T-1} \sim b)}$$

$$= \frac{\prod_{k=t}^{T-1} \pi(A_k \mid S_k) \cancel{p(S_{k+1} \mid S_k, A_k)}}{\prod_{k=t}^{T-1} b(A_k \mid S_k) \cancel{p(S_{k+1} \mid S_k, A_k)}}$$

$$= \prod_{k=t}^{T-1} \left(\frac{\pi(A_k \mid S_k)}{b(A_k \mid S_k)} \right)$$

Recall: We wish to find values under target policy π but we have available estimates of values under behavior policy b

$$E[G_t \mid S_t = s, b] = v_b(s)$$

However, we want $v_\pi(s)$.

Consider

$$E[\rho_{t:T-1} G_t \mid S_t = s, b]$$

$$= E\left[\prod_{k=t}^{T-1} \frac{\pi(A_k \mid S_k)}{b(A_k \mid S_k)} G_t \,\middle|\, S_t = s, b\right]$$

$$= v_\pi(s)$$

Idea:

Given 2 distributions

p_π & p_b

We want to find

$E_\pi(x)$ but we have

are samples from

p_b.

Let $\rho(x) = \dfrac{p_\pi(x)}{p_b(x)}$

$$E_b[\rho X] = \sum_x \cancel{p_b(x)} \cdot \frac{p_\pi(x)}{\cancel{p_b(x)}} \cdot x$$

$$= \sum_x x \, p_\pi(x)$$

$$= E_\pi[X]$$

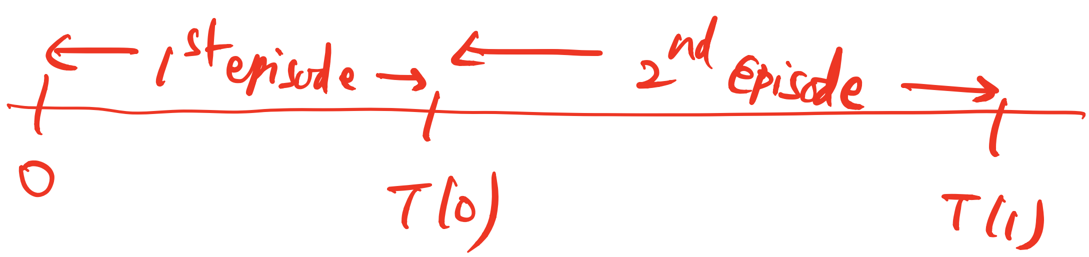
Thus, $E_b[\rho X] = E_\pi[X]$

Monte-Carlo Algorithm — to estimate $V_\pi(s)$ using estimates from b

Let, $\mathcal{T}(s) \equiv$ set of all time steps

in which state s is visited
(every visit method)

- $\mathcal{T}(s) \equiv$ set of all time steps in which state s is visited the first time in an episode (first visit method)



$T(t)$ = first time of termination following time t.

Now $\{G_t\}_{t \in \mathcal{T}(s)}$ are the returns

pertaining to state s and $\{\rho_{t:T(t)-1}\}_{t \in \mathcal{T}(s)}$

are the corresponding IS ratios.

Regular M.C Estimate:

$$V(s) \triangleq \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{|\mathcal{T}(s)|}$$

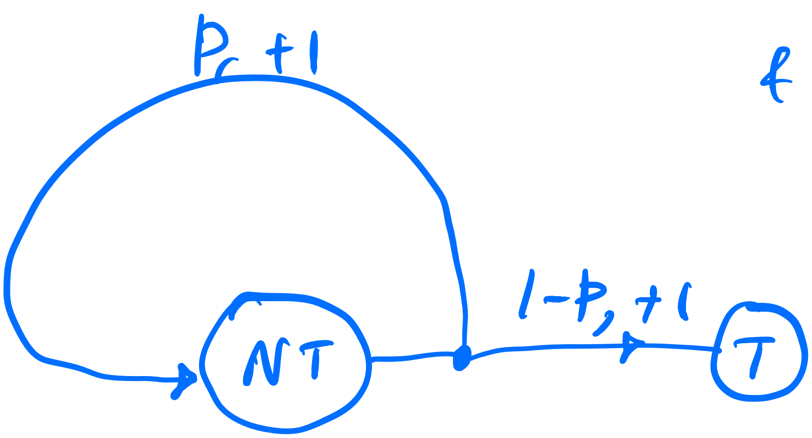
Weighted M.C Estimate:

$$V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1}}$$

<u>Exercise</u> : 2 state MDP — NT, T

NT → nonterminal, T → terminal

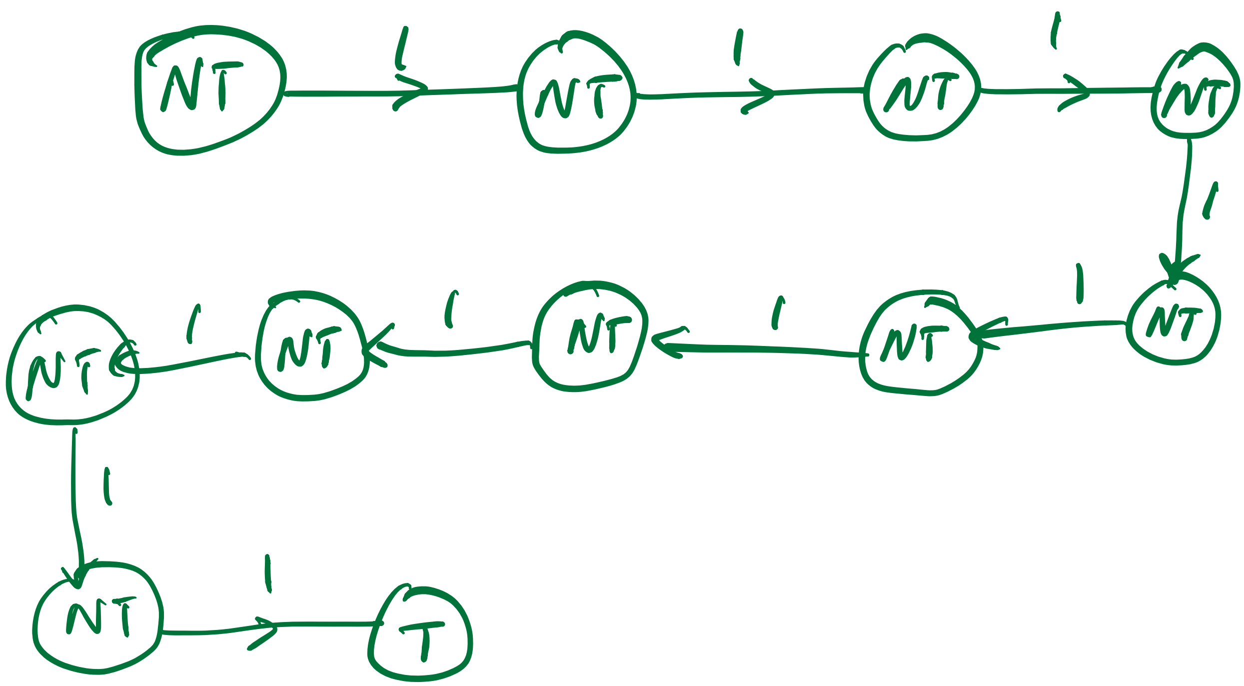
& single action



Let $\gamma = 1$.

Consider an episode that lasts for 10 steps with a return of 10.

Q) What are the first-visit and every-visit estimators of the value of the nonterminal state?



Return on the episode = 10.

First visit method:

Av. return = 10.
(on the episode)

Every visit method

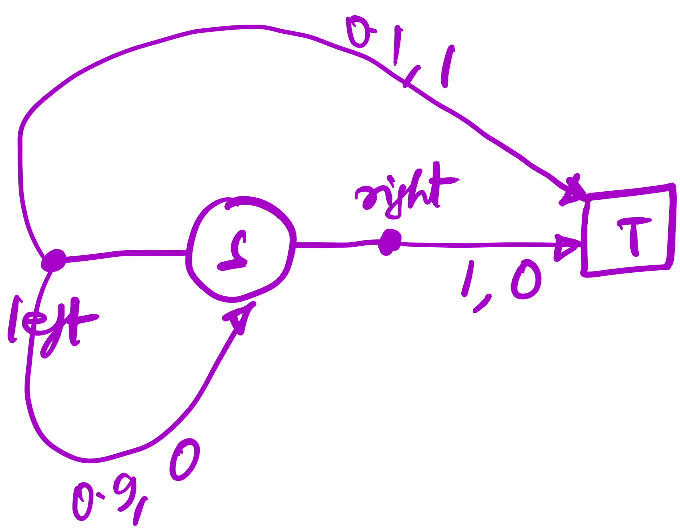
Returns: 10, 9, 8, 7, 6, 5, 4, 3,
2, 1

Av. Return

$$= \frac{1}{10}\left(10+9+8+7+6+5+4+3+2+1\right)$$

$$= \frac{1}{\cancel{10}_{2}} * \frac{\cancel{10}^{5} * 11}{\cancel{2}} = \frac{11}{2} = 5.5.$$

Example to illustrate that ordinary IS can give ∞ variance.



Target policy: $\pi(\text{left}|s) = 1$

Behavior policy: $b(\text{left}|s) = \frac{1}{2}$

$b(\text{right}|s) = \frac{1}{2}$

We know that

$$\mathrm{Var}(x) = E[x^2] - E[x]^2$$

$$E[x]$$

$$= \ ?$$

Consider

$$E_b\left[\prod_{t=0}^{T-1} \frac{\pi(A_t|S_t)}{b(A_t|S_t)} G_0\right]$$

$$= \frac{1}{2} * 0.1 * \frac{1}{0.5} * 1 + \overbrace{\frac{1}{2} * 1 * \frac{1}{0.5} * 0}^{=0}$$

$$= 0.1$$

[length 1 episode]

$$+ \frac{1}{2} * 0.9 * \frac{1}{2} * 0.1 * \frac{1}{0.5} * \frac{1}{0.5}$$

$$= 0.9 * 0.1$$ [length 2 episode]

$$+ \frac{1}{2} * 0.9 * \frac{1}{2} * 0.9 * \frac{1}{2} * 0.1 * \frac{1}{0.5} * \frac{1}{0.5} * \frac{1}{0.5}$$

$$= (0.9)^2 * 0.1$$ [length 3 episode]

$$+ \cdots$$

$$= (0.1) + 0.9 * 0.1 + (0.9)^2 * 0.1 + \cdots$$

$$= 0.1 \left[1 + 0.9 + (0.9)^2 + \cdots\right]$$

$$= 0.1 * \frac{1}{1-0.9} = 1$$

For finding variance, let's compute the second moment

$$E_b\left[\left(\prod_{t=0}^{T-1} \frac{\pi(A_t|S_t)}{b(A_t|S_t)} G_0\right)^2\right]$$

$$= \frac{1}{2} * 0.1 * \left(\frac{1}{0.5}\right)^2 \quad [\text{length 1 episode}]$$

$$+ \frac{1}{2} * 0.9 * \frac{1}{2} * 0.1 * \left(\frac{1}{0.5} * \frac{1}{0.5}\right)^2$$

$$[\text{length 2 episode}]$$

$$+ \frac{1}{2} * 0.9 * \frac{1}{2} * 0.9 * \frac{1}{2} * 0.1 * \left(\frac{1}{0.5} * \frac{1}{0.5} * \frac{1}{0.5}\right)^2$$

[length 3 episode]

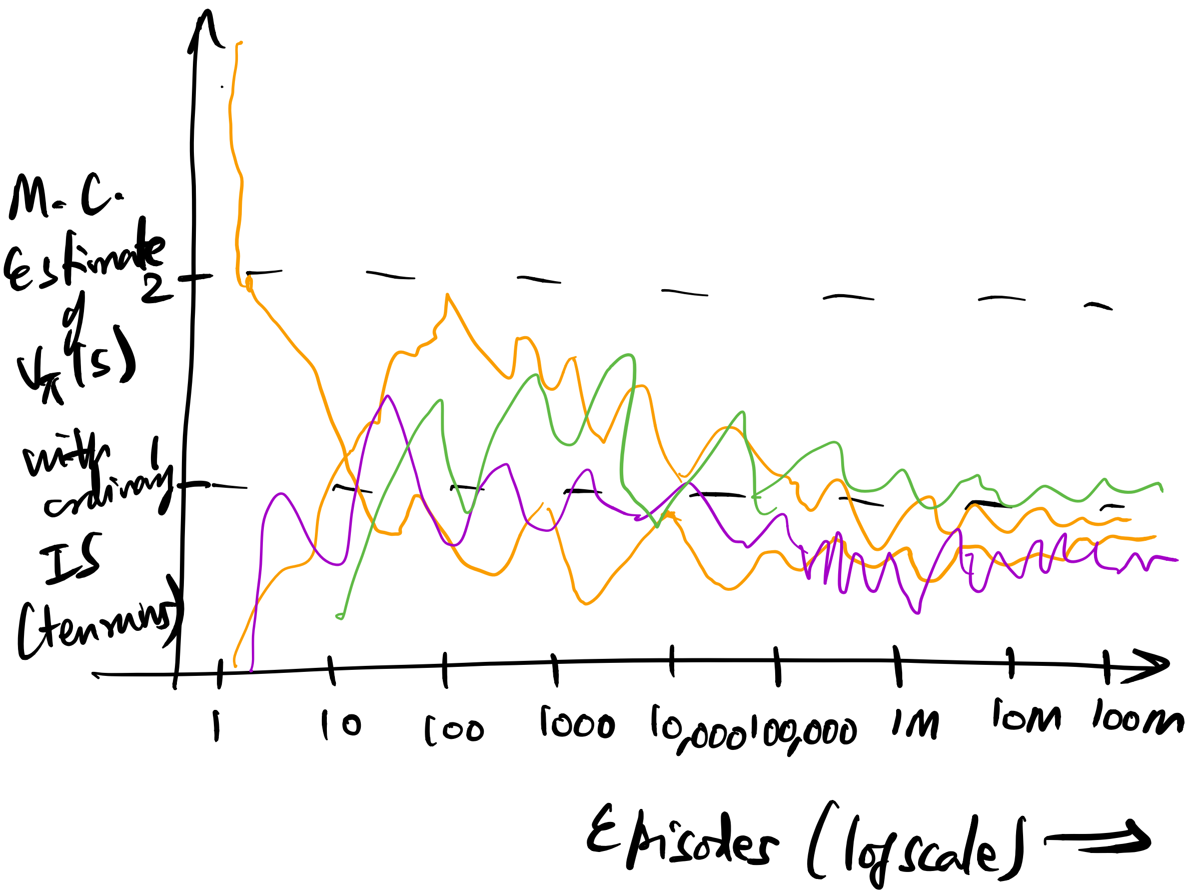
$+ \cdots$

$$= 0.1 * \sum_{k=0}^{\infty} (0.9)^k \, 2^k * 2$$

$$= 0.2 * \sum_{k=0}^{\infty} (1.8)^k$$

$$= \infty$$

Weighted IS algorithm produces weighted average of only returns consistent with target policy & all these would be exactly 1.



Incremental Implementation

Suppose the sequence of returns is $G_1, G_2, \dots, G_{n-1}$ all starting in same state & G_i has weight W_i

(e.g., $W_i = \rho_{t_i : T(t_i)-1}$.

Let
$$V_n = \frac{\sum_{k=1}^{n} W_k G_k}{\sum_{k=1}^{n} W_k}, \quad n \geq 1$$

Then
$$V_{n+1} = \frac{\sum_{k=1}^{n} W_k G_k + W_{n+1} G_{n+1}}{\sum_{k=1}^{n+1} W_k}$$

$$= \frac{\sum_{k=1}^{n} W_k}{\sum_{k=1}^{n+1} W_k} \left(\frac{\sum_{k=1}^{n} W_k G_k}{\sum_{k=1}^{n} W_k} \right)$$

$$+ \frac{W_{n+1}}{\sum_{k=1}^{n+1} W_k} G_{n+1}$$

$$= \frac{\sum_{k=1}^{n} W_k}{\sum_{k=1}^{n+1} W_k} V_n + \frac{W_{n+1}}{\sum_{k=1}^{n+1} W_k} G_{n+1}$$

$$= V_n + \frac{W_{n+1}}{\sum_{k=1}^{n+1} W_k} \left(G_{n+1} - V_n \right)$$

Start with an initial estimate V_0 of value function.

Update as

$$V_1 = V_0 + \frac{W_1}{W_0 + W_1}(G_1 - V_0)$$

Incremental Update Algorithm:

$$\boxed{\begin{aligned} V_{n+1} &= V_n + \frac{W_{n+1}}{C_{n+1}}\left[G_{n+1} - V_n\right], \quad n \geq 0 \\ C_{n+1} &= C_n + W_{n+1} \end{aligned}}$$

with $C_0 = 0$.

Also, V_0 is arbitrary chosen

This algorithm also works for on-policy case, by choosing $\pi = b$.